Variant calling

SNPs and indels

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Learning outcomes

- **Calling single nucleotide polymorphisms (SNPs) and insertions/deletions (indels)**
  - tools
  - considerations
- **Understanding vcf format**

Identifying variation within genomes

- Identifying variants is important for evolution studies, interring resistance mutations and in population genomics studies
  - Reads aligned to a reference genome enables identification of differences between individuals
- SNPs are the most abundant variants, along with insertion/deletions
- Widely used software for variant calling includes GATK and Freebayes
  - Can call variants individually or across multiple samples
  - Determine genotypes e.g. at each position, determine whether variant is homozygous reference, heterozygous or homozygous alternate
- Creates variant call format (vcf) file

Variant call format (VCF)

The details

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS      ID         REF   ALT    QUAL  FILTER   INFO                              FORMAT       NA00001         NA00002
NA00003
20     14370    rs6054257  G     A      29    PASS    NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ  0|0:48:1:51,51
1|0:48:8:51,51   1/1:43:5:.,.
20     17330    .          T     A      3     q10     NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ  0|0:49:3:58,50
0|1:3:5:65,3     0/0:41:3
20     1110696  rs6040355  A     G,T    67    PASS    NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ  1|2:21:6:23,27
2|1:2:0:18,2     2/2:35:4
20     1230237  .          T     .      47    PASS    NS=3;DP=13;AA=T                   GT:GQ:DP:HQ  0|0:54:7:56,60
0|0:48:4:51,51   0/0:61:2
20     1234567  microsat1  GTC   G,GTCT 50    PASS    NS=3;DP=9;AA=G                    GT:GQ:DP     0/1:35:4        0/2:17:2
1/1:40:3
```

Calling high-quality variants

Commands using GATK

```
gatk HaplotypeCaller –R reference.fa –I isolate_name.post_recal_reads.bam –O
isolate_name.raw_variants_recal.vcf -ERC GVCF –-pcr-indel-model NONE –ploidy 1
–stand_call_conf 30 –mbq 20 –A QualByDepth

gatk GenotypeGVCFs –R reference.fa -V isolate_name.raw_variants_recal.vcf -O
isolate_name.genotyped_variants_recal.vcf
```

Calling variants

Considerations

- Repetitive regions
  - Most fungi have 1-25% repetitive DNA content and genome size is correlated with number of transposon families hosted within the genome
  - Difficult to call SNPs in repetitive regions
    - lower quality scores in repetitive regions, so variant calls may be false positives
  - RepeatMasker to mask regions of repeats in the reference genome
    - Include in HaplotypeCaller using the -XL parameter